

Passage Qualification

Turn two healthy boundaries into one bounded record

Dwell first, decide once, suppress duplicates

Lesson	035 — component	Time	120–180 minutes
Prerequisites	Lessons 032 and 034	Board	Mega 2560 evidence panel
Evidence	TP-A, TP-B, three LED witnesses	Status	host verified; bench open
Energy class	E0 software fixture; no sensor connection or actuation		

Scope gate. `PassageQualifier` is pure software. It owns no pin, clock, endpoint, storage, callback, sensor, motor, gate, or lock. TP-A and TP-B in the canonical sketch are deterministic fixture frames, not electrical inputs. Do not connect a magnetic-family specimen until Lesson 034's exact-revision qualification and schematic gates are complete.

1 Mission and evidence chain

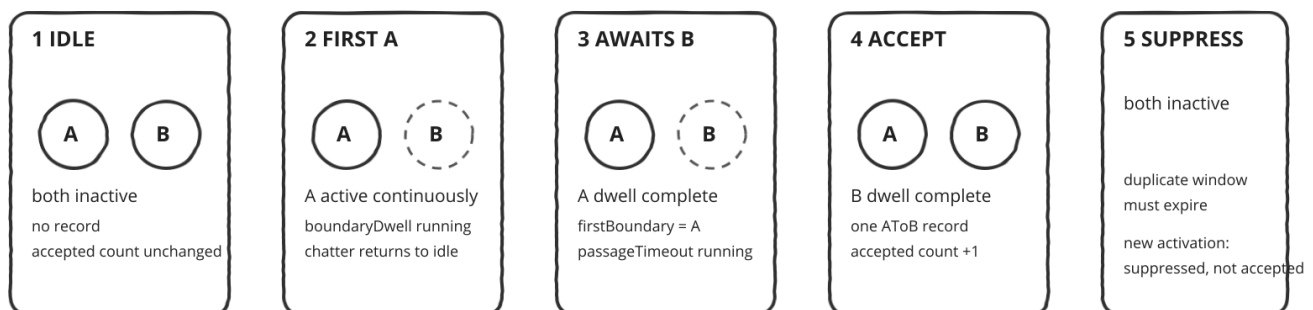
The lesson preserves:

two healthy observations → qualified boundary order → one passage record.

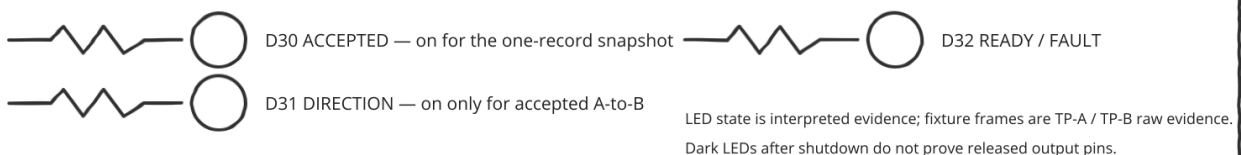
By the end, you can explain why a passage needs continuous dwell at two geometric boundaries; distinguish accepted, timed-out, ambiguous, suppressed, and faulty evidence; and prove that optional position corroborates direction without becoming passage authority.

2 Build the visible story in stages

DEPENDENT VISIBLE STAGES — SOFTWARE FIXTURE, E0



EVIDENCE PANEL — THREE SEPARATE 1 kOhm LED PATHS



Alternative text: five pencil stages show Idle, boundary A dwell, AwaitingSecond, one accepted A-to-B record, and duplicate suppression; below, D30 shows accepted, D31 distinguishes accepted A-to-B, and D32 shows ready versus fault through separate 1 k Ω LED paths.

Run the canonical software fixture with only the three resistor-limited LEDs. First prove D32 ready. Then watch one A-to-B record light D30 and D31. The later B-to-A record lights D30 without D31. Qualifier timestamps advance in 10 ms steps while each frame remains board-visible for 250 ms. This makes the direction contrast human-readable without changing qualification time or using Serial or an unidentified sensor.

Evidence	Provisional panel role	Interpretation
TP-A, TP-B	internal fixture trace labels	copied boundary activity and time; not board pins
D30	LED through 1 k Ω	latest snapshot contains Accepted record
D31	LED through 1 k Ω	that accepted record is A-to-B
D32	LED through 1 k Ω	qualifier is not in Fault

These roles are not a sensor schematic. Each direct LED is bounded to at most 5 mA at 5 V, but darkness does not prove output-pin release. The sketch stops and releases the evidence panel independently after 120 seconds.

3 Public contract

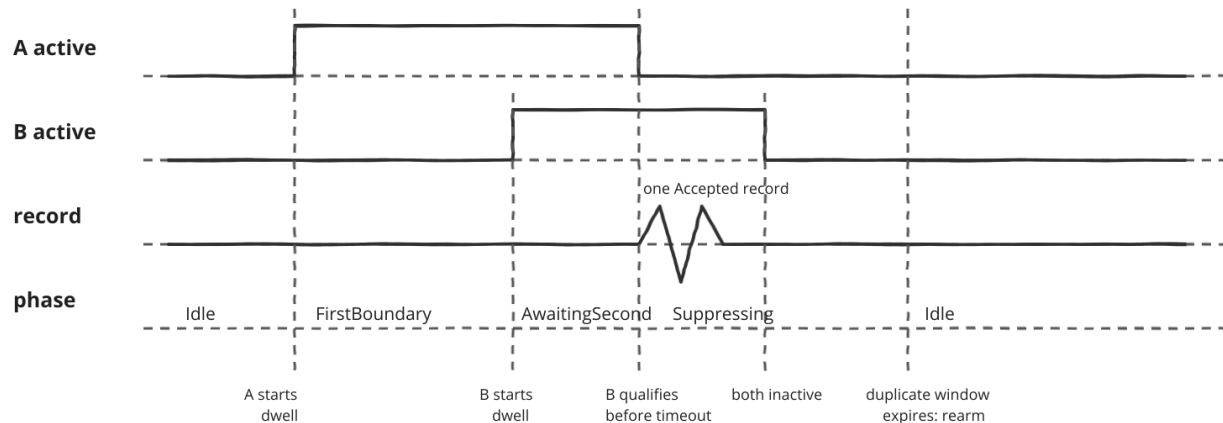
`PassageQualifierConfig` contains nonzero `boundaryDwell`, `passageTimeout`, and `duplicateWindow`; time-out is at least dwell and every duration is below the unsigned half-range. `PassageInput` carries one timestamp, two `MagneticObservation` values, and optional copied position evidence.

Snapshot or record field	Meaning
<code>phase</code> , <code>firstBoundary</code> , <code>elapsed</code>	current qualification state
<code>hasRecord</code> , <code>record.sequence</code>	one terminal record in this snapshot
<code>direction</code> , <code>disposition</code>	AToB/BToA and accepted/rejected reason
<code>onset</code> , <code>end</code>	retained qualification timestamps
<code>position</code>	copied onset/end/delta plus reliability and saturation
<code>acceptedCount</code> , <code>suppressedCount</code>	saturating evidence totals
<code>status</code>	lifecycle, time, or qualification result

`initialize()` validates policy; `update(input)` advances once; `snapshot()` observes without consuming; and `reset()` returns to the just-initialized state and clears history and counts. A record remains for exactly the terminal snapshot and disappears on the next later update.

4 Timing, direction, and suppression

ACCEPTED A-TO-B TRACE — EXACT EDGES ARE PART OF THE CONTRACT



OTHER TERMINAL PATHS

late second dwell = TimedOut

both qualify together = Ambiguous

first retreats = wait for both inactive

bad evidence/time = Fault

Alternative text: pencil timing tracks show A qualifying first, B qualifying before the timeout edge, one accepted A-to-B record, and rearm only after both boundaries are inactive and the duplicate window expires; side cases name timeout, ambiguity, retreat, and fault.

The timeout begins when the first boundary finishes dwell. The opposite boundary may finish dwell at the exact timeout edge; a later completion times out. Acceptance enters **Suppressing**. Rearm requires both inactive and duplicate-window expiry. An activation inside that window emits one **DuplicateSuppressed** record and increments only the suppressed count; holding it active cannot count it again.

If the first boundary retreats before the second qualifies, the event cannot reverse into acceptance. The qualifier waits until both are inactive, then returns to **Idle** without a record. Simultaneous qualification is **Ambiguous**.

5 Fault precedence and recovery

Each input timestamp must equal both observation timestamps. Only observations with **Ok** status and **Valid** quality participate. Precedence is invalid time, source fault, ambiguity, timeout, then completion. An active candidate emits one **EvidenceFault** record before entering **Fault**; an idle source fault enters **Fault** without inventing a record. Recovery requires a later healthy both-inactive frame and emits nothing.

Natural 32-bit rollover is valid. An exact half-range or larger apparent jump faults without partial mutation. Repeating an identical same-time frame is idempotent; changing any same-time evidence faults.

Optional position is corroboration only. A nonzero, unsaturated delta is reliable when its sign agrees with A-to-B positive and B-to-A negative. Missing, zero, opposite, source-failed, or saturated position remains explicit but does not reject an otherwise accepted passage.

6 Narrative Mega example

```
void loop ()
```

```

{
    observeFixture ();

    if (!decidePassage () || !actuateEvidence ())
    {
        stopSafely ();
    }
}

```

The full sketch acquires the three witnesses, configures the qualifier, starts the bounded fixture, and keeps observe–decide–actuate order. It creates **MagneticObservation** fixture values in software and never reads a physical sensor.

7 Predict, observe, interpret

1. Predict D32 after successful initialization. Observe it separately from D30/D31; interpret it only as qualifier/panel readiness.
2. Predict the first A dwell, later B dwell, direction, and exact record snapshot. Inspect the TP-A/TP-B fixture-trace columns (they are not board pins), then observe the board-visible D30/D31 pair.
3. Reverse the trace. Predict D30 on and D31 off for accepted B-to-A.
4. Place an activation inside suppression. Predict one suppressed record, no D30 accepted indication, and no accepted-count change.
5. Inject timestamp mismatch in the host fixture. Predict Fault, then a silent recovery only on a later healthy both-inactive frame.
6. Invoke shutdown. Predict dark LEDs, then verify released output state separately; LED darkness alone is insufficient.

8 Diagnosis and stop conditions

Observation	Interpret and next check
D30 never pulses	inspect dwell, timeout, health, and terminal snapshot timing
D31 disagrees	compare firstBoundary and accepted direction
repeated accepted pulses	inspect both-inactive and duplicate-window frames
Fault persists	supply a later healthy both-inactive frame
position unreliable	inspect presence, source status, sign, zero, and saturation
LED dark after stop	measure release; do not infer high impedance from darkness

Stop rule. This lesson authorizes only the software fixture and resistor-limited LED panel. Remove USB for heat, odor, reset, unstable rail, unexpected LED state, or current-limit trip. Reconfigure only while unpowered. Do not connect unknown specimens, motion, a gate, lock, vehicle, launcher, ignition path, or safety system.

9 Exercises and software evidence

1. Draw the B-to-A counterpart of the accepted trace.

2. Explain why retreat cannot become reverse acceptance.
3. Order ambiguity, timeout, and completion at one timestamp.
4. Explain why an unreliable position delta does not reject a passage.
5. Give a rollover trace that completes boundary dwell across zero.

Host tests cover both directions, exact dwell/timeout/suppression edges, chatter, retreat, ambiguity, faults, recovery, same-time inputs, deterministic replay, rollover, position reliability/saturation, reset, and saturating sequences and counters. Compilation, tests, this PDF, and LED behavior are not physical sensor acceptance. The searchable HTML lesson is the primary accessible route. This PDF remains untagged and makes no PDF/UA claim.